

Non-small cell lung cancer

Adenocarcinoma (TTF-1, napsin A, CK7) · squamous (CK5/6, p63) · large cell · **9th edition TNM** (Jan 2025) — new N2a/N2b and M1c1/M1c2 subdivisions

Stanford Oncology Handbook 2025, NSCLC (Salter A, Roy M) · NCCN — the chapter stresses the targeted landscape moves fast; re-check NCCN at diagnosis

WORKUP

- Chest CT **with contrast** (nodes need contrast)
- **Brain MRI with contrast for every stage except IA**
- PET/CT for distant disease (or CT C/A/P + bone scan)
- Biopsy: EBUS, CT-guided transthoracic, met site, or pleural cytology — **non-decalcified cores** preferred for molecular
- **Pathologic mediastinal nodal evaluation** for non-metastatic disease, especially stage II–III
- **PD-L1 in ALL cases.** Molecular sequencing in all adenocarcinoma; add for squamous if never/light smoker or small biopsy
- Stage IIIB or IV: **concurrent tissue + liquid** molecular testing (ctDNA ~7-day turnaround)

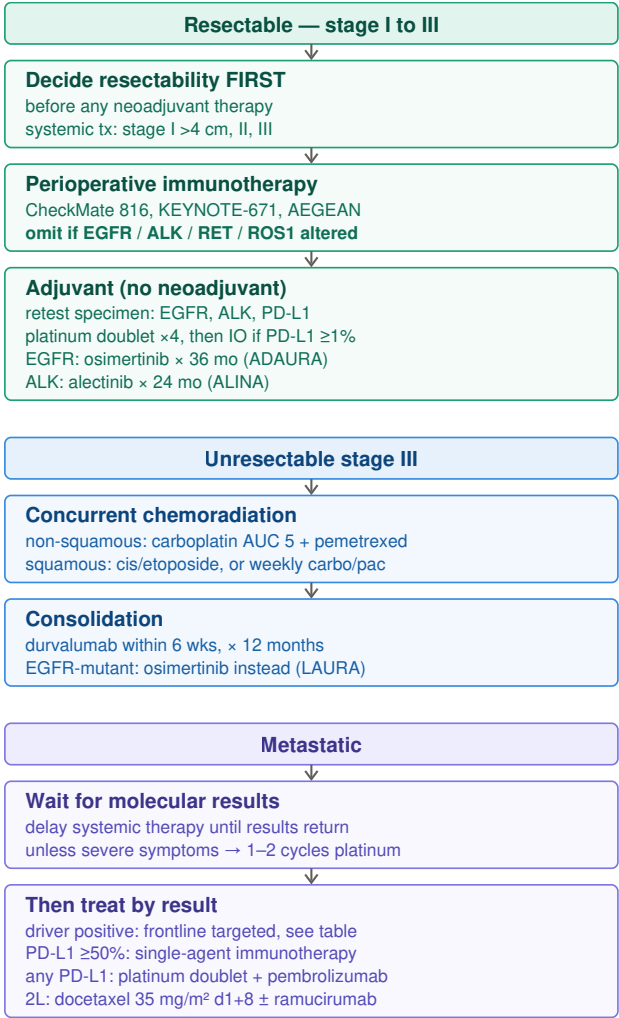
STAGING — TNM 9TH

T1 / T2	≤3 cm / >3–5 cm or visceral pleura, main bronchus
T3 / T4	>5–7 cm or chest wall / >7 cm or mediastinal invasion
N1 / N2 / N3	ipsilateral hilar / ipsilateral mediastinal — N2a single station, N2b multiple / contralateral or supraclavicular
M1a / M1b	pleural-pericardial or contralateral nodule / single extrathoracic met
M1c1 / M1c2	multiple mets, single organ system / multiple systems

KEY TRIALS

CheckMate 816	neoadjuvant platinum + nivolumab ×3 (NEJM 2022)
KEYNOTE-671	perioperative pembrolizumab (NEJM 2023)
ADAURA / ALINA	adjuvant osimertinib ×36 mo / alectinib ×24 mo
PACIFIC	consolidation durvalumab after chemoRT (NEJM 2018)
LAURA	consolidation osimertinib, EGFR-mutant stage III (2024)
FLAURA / FLAURA2	osimertinib ± chemo, 1L EGFR-mutant
CROWN	lorlatinib 1L ALK-positive (2024 update)

TREATMENT PATHWAY



BIOMARKER → THERAPY

Alteration	Therapy
EGFR 19del / L858R	Osimertinib 80 mg daily (FLAURA); or + carbo/pemetrexed ×4 (FLAURA2); or amivantamab + lazertinib (MARIPOSA)
— adjuvant	Osimertinib 80 mg × 36 months (ADAURA) — omit IO
— stage III post-chemoRT	Consolidation osimertinib until progression (LAURA)
ALK rearrangement	Lorlatinib 100 mg daily (CROWN); adjuvant alectinib 600 BID × 24 mo (ALINA) — omit IO
ROS1 / BRAF V600E / MET ex14 / RET	Frontline targeted therapy — agents change fast, check NCCN. Omit IO in RET-rearranged resectable disease
PD-L1 ≥50%, no driver	Pembrolizumab 200 q3wk (or 400 q6wk) · atezolizumab 1200 q3wk · cemiplimab 350 q3wk
PD-L1 ≥1%, no driver	Ipilimumab 1 mg/kg + nivolumab 3 mg/kg q6wk (CheckMate 227)
PD-L1 <1%	Chemo + dual IO (durvalumab/tremelimumab or ipi/nivo) — especially squamous

REGIMENS & DOSING

Regimen	Dosing	Cycle
Platinum doublet	cisplatin 75 d1 + pemetrexed 500 (non-squamous) or gemcitabine 1250 d1+8 or docetaxel 75	q21d ×3–4
+ immunotherapy	nivolumab 360 · pembrolizumab 200 · durvalumab 1500	q21d
ChemoRT non-squamous	carboplatin AUC 5 + pemetrexed 500	q21d ×4
ChemoRT squamous	cisplatin 50 d1,8,29,36 + etoposide 50 d1–5, 29–33; or carbo AUC 2 + paclitaxel 45 weekly	with daily RT
Durvalumab consolidation	10 mg/kg q14d × 12 months (or 1500 mg q4wk)	q14d
Docetaxel (2L, preferred)	35 mg/m² d1+8 split dose — may reduce to 30 or 25	q21d
Docetaxel + ramucirumab	docetaxel 75 d1 + ramucirumab d1 (prefer split-dose docetaxel)	q21d
Pemetrexed maintenance	500 mg/m² d1 — non-squamous only	q21d

All doses mg/m² unless stated. **Pemetrexed is not used in squamous.** Bevacizumab is contraindicated in squamous, significant hemoptysis, or large cavitary tumors.

MONITORING & CAUTIONS

- **Baseline HBsAg + HbCAb** before any immunosuppressive therapy
- **Caution with targeted agents — especially osimertinib and crizotinib — within 3 months after immunotherapy** (increased toxicity); IO is also less effective in EGFR/ALK-driven disease
- **Osimertinib:** echo at baseline and q3 mo × 6 mo; EKG at baseline, 2 wk, 6 wk
- **Alectinib:** CK at baseline, 2 wk, then q3 mo · LFTs q2 wk × 3 mo. **Lorlatinib:** CNS effects (seizures, mood, cognition) + hyperlipidemia (median onset 15 days)
- **Amivantamab + lazertinib:** VTE prophylaxis for the first 4 months
- **Pemetrexed:** folic acid from ≥1 wk before · B12 q9 wk · dexamethasone for rash